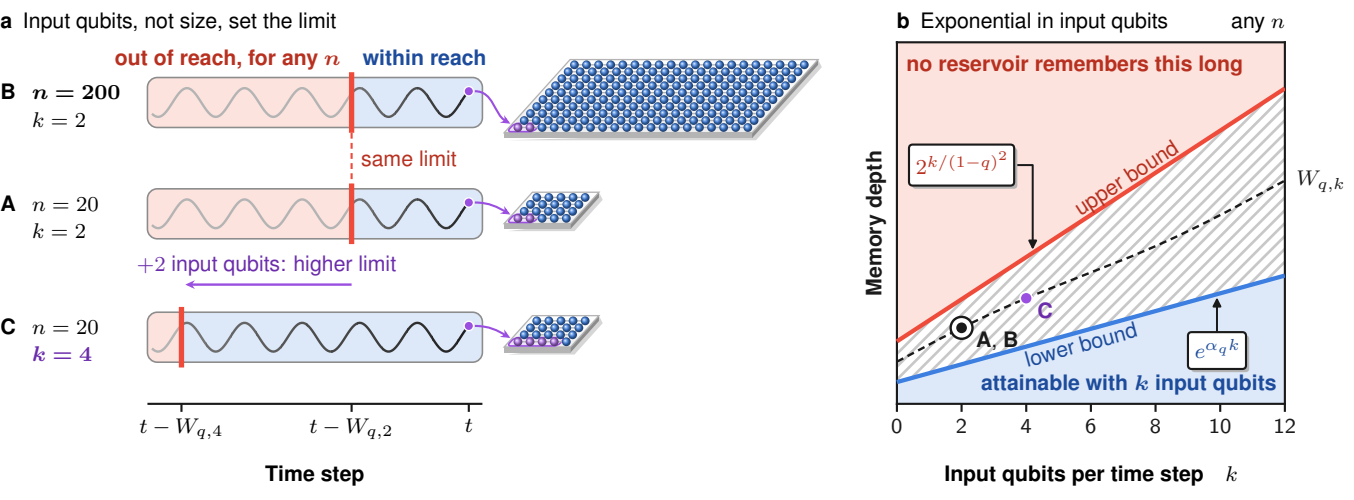

a Input qubits, not size, set the limit
out of reach, for any n
within reach
B $n = 200$
$k = 2$
same limit
A $n = 20$
$k = 2$
+2 input qubits: higher limit
C $n = 20$
$k = 4$
$t - W_{q,4}$
$t - W_{q,2}$
t
Time step
b Exponential in input qubits
any n
no reservoir remembers this long
$2^{k/(1-q)^2}$
upper bound
$W_{q,k}$
Memory depth
C
A, B
lower bound
$e^{\alpha_q k}$
attainable with k input qubits
0
2
4
6
8
10
12
Input qubits per time step k